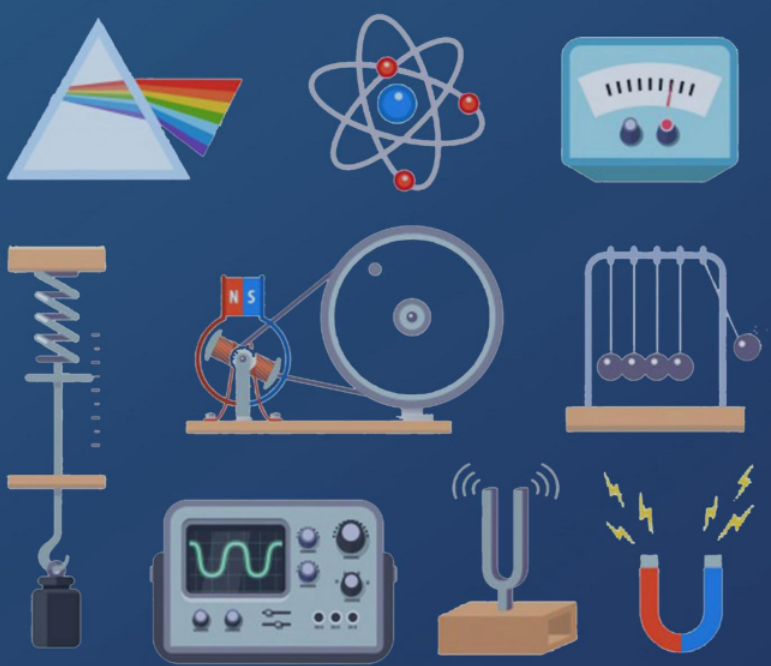
PHYSICS

Newton Laws of Motion



SHEET

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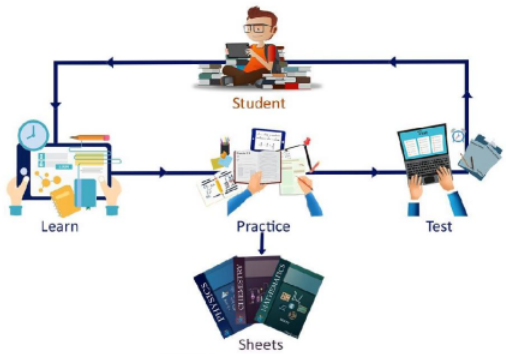
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 5
5.	Exercise 2	Page - 9
6.	Exercise 2A	Page - 11
7.	Answer Key	Page - 15

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Newton's Laws of Motion	1-3	1		
2	Different Types of Forces		2		
3	Translational Motion	4-8	3, 5, 4, 11	1, 2	6,7
4	Translational Equilibrium	9-14	6,9	3, 4, 5	
5	Pulley, Weighing Machine and Heavy String	15-18	7	6, 7, 8	1,2,8-11
6	Spring Force	19-23	12	9	3,12-17
7	Pseudo Force	24	10		
8	Constraint Motion	25-26	8	10,11,12	4
9	Wedge Constraint	27			5,18

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Newton's Laws of Motion

Q1. A man getting down a running bus, falls forward because-

(A) due to inertia of rest, road is left behind and man reaches forward

(B) due to inertia of motion upper part of body continues to be in motion in forward direction while feet come to rest as soon as they touch the road

(C) he leans forward as a matter of habit

(D) of the combined effect of all the three factors stated in (1), (2) and (3)

Q2. A vehicle of 100 kg is moving with a velocity of 5 m/sec. To stop it in $\frac{1}{10}$ s. The required force in opposite direction is-

(A) 5000 newton (B) 500 newton

(C) 50 newton (D) 1000 newton

Q3. Two astronauts A and B connected with a rope stay stationary in free space relative to their spaceship. Mass of A is more than that of B and the rope is straight. Astronaut A starts pulling the rope but astronaut B does not. If you were the third astronaut in the spaceship, what do you observe?

(A) Astronaut B accelerates towards A and A remains stationery.

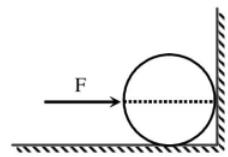
(B) Both accelerate towards each other with equal accelerations of equal modulus.

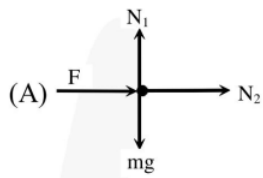
(C) Both accelerate towards each other but acceleration of B is greater than that of A.

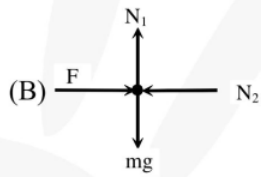
(D) Both accelerate towards each other but acceleration of B is smaller than that of A.

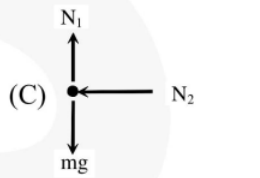
Translational Motion

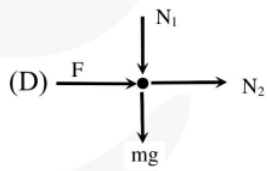
Q4. A ball of mass m kept at the corner as shown in the figure, is acted by a horizontal force F. The correct free body diagram of ball is -



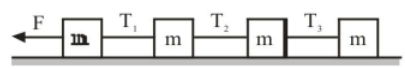
(A)


(B)


(C)


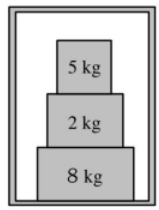
(D)


Q5. Four blocks of same mass connected by cords are pulled by force F on a smooth horizontal surface, as in figure. The tension T_1, T_2 and T_3 will be -



(A) $T_1 = F/4, T_2 = 3F/2, T_3 = F/4$
(B) $T_1 = F/4, T_2 = F/2, T_3 = F/2$
(C) $T_1 = 3F/4, T_2 = F/2, T_3 = F/4$
(D) $T_1 = 3F/4, T_2 = F/2, T_3 = F/2$

Q6. Three boxes are placed in a lift. When acceleration of the lift is 4 m/s^2, the net force on the 8 kg box is closest to



(A) 80 N

(B) 48 N

(C) 40 N

(D) 32 N

Q7. A block of mass m is placed in contact with one end of a smooth tube of mass M (see figure). A horizontal force F acts on the tube in each case (i) and (ii). Then mark **incorrect** option :



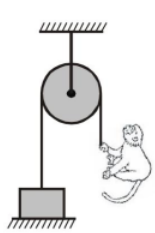
(A) $a_m = 0$ and $a_M = \frac{F}{M}$ in (i)

(B) $a_m = a_M = \frac{F}{M+m}$ in (i)

(C) $a_m = a_M = \frac{F}{M+m}$ in (ii)

(D) Net force on m is $\frac{mF}{M+m}$ in (ii)

Q8. A monkey weighing 10 kg is climbing up a light rope and frictionless pulley attached to 15 kg mass at other end as in figure. In order to raise the 15 kg mass off the ground the monkey must climb-up -



(A) with constant acceleration g/3.

(B) with an acceleration greater than g/2.

(C) with an acceleration greater than g/4.

(D) It is not possible because weight of monkey lesser than the block.

Translational Equilibrium

Q9. Under what condition(s) will an object be in equilibrium ?

(A) Only if it is at rest

(B) Only if it is moving with constant velocity

(C) Only if it is moving with constant acceleration

(D) If it is either at rest or moving with constant velocity

Q10. If the net force on a body is zero, will it definitely be at rest ?

(A) Yes

(B) Not necessary. Net force is zero on the body moving with accelerating velocity

(C) Not necessary. Net force is zero on the body moving with decelerating velocity

(D) Not necessary. This is possible that the body is moving with constant velocity

Q11. Ten coins are placed on top of each other on a horizontal table. If the mass of each coin is 10 gm and acceleration due to gravity is 10 m/s^2, what is the magnitude and direction of the normal force on the 7th coin applied by 8th coin (counted from the bottom) ?

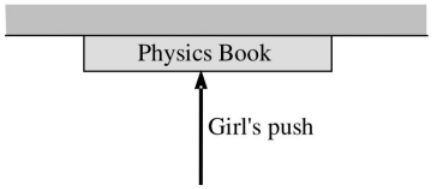
(A) 0.1 N vertically downwards

(B) 0.7 N vertically upwards

(C) 0.3 N vertically downwards

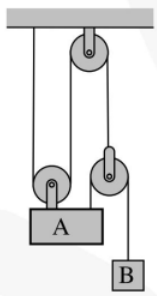
(D) 0.3 N vertically upwards

Q12. A girl pushes her physics book up against the horizontal ceiling of her room as shown in the figure. The book weighs 20 N and she pushes upwards with a force of 25 N. The choices below list the magnitudes of the contact force F_{CB} between the ceiling and the book, and F_{BH} between the book and her hand. Select the correct pair.



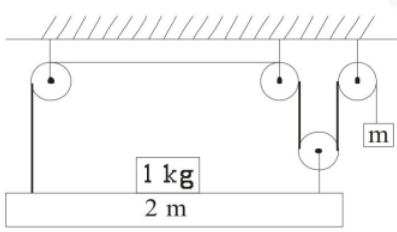
(A) $F_{CB} = 20$ N and $F_{BH} = 25$ N
(B) $F_{CB} = 25$ N and $F_{BH} = 45$ N
(C) $F_{CB} = 5$ N and $F_{BH} = 25$ N
(D) $F_{CB} = 5$ N and $F_{BH} = 45$ N

Q13. In arrangement shown the block A of mass 15 kg is supported in equilibrium by the block B. Mass of the block B is closest to



(A) 2 kg (B) 3 kg
(C) 4 kg (D) 5 kg

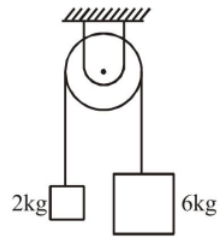
Q14. If the arrangement shown in the figure is in equilibrium, mass m of the block is closest to -



(A) 1.0 kg (B) 1.5 kg
(C) 3.0 kg (D) 6.0 kg

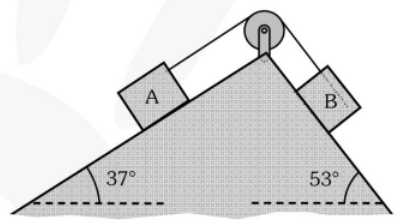
Pulley, Weighing Machine and Heavy String

Q15. Find acceleration of blocks :-



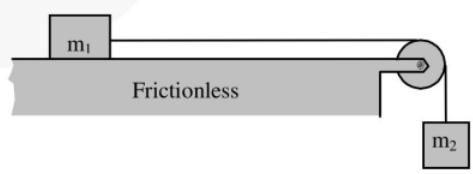
(A) g (B) g/2
(C) g/3 (D) g/4

Q16. The acceleration of the block A is 2 m/s^2 down the incline. The ratio of the mass of A to that of B is -



(A) 7 : 6 (B) 5 : 2
(C) 5 : 3 (D) 3 : 4

Q17. In the arrangement shown, the blocks of unequal masses are held at rest. When released, acceleration of the blocks is -



(A) g/2.
(B) g.
(C) a value between zero and g.
(D) a value that could be greater than g.

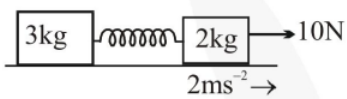
Q18. A man is standing on a weighing machine with a block in his hand. The machine records w. When he takes the block upwards with some acceleration the machine records w_1. When he takes the block down with same acceleration, the machine records w_2.Then choose correct option :

(A) $w_1 = w = w_2$ (B) $w_1 < w < w_2$

(C) $w_2 < w < w_1$ (D) $w_2 = w_1 > w$

Spring Force

Q19. Find the acceleration of 3 kg mass when acceleration of 2 kg mass is 2 ms^{-2} as shown in figure.



(A) 3 ms^{-2} (B) 2 ms^{-2}

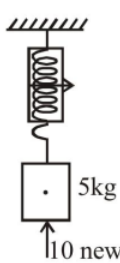
(C) 0.5 ms^{-2} (D) zero

Q20. An elastic spring of relaxed length ℓ_o and force constant k is cut into two parts of lengths ℓ_1 and ℓ_2. The force constants of these parts are respectively.

(A) $\frac{k\ell_o}{\ell_1}$ and $\frac{k\ell_o}{\ell_2}$ (B) $\frac{k\ell_1}{\ell_0}$ and $\frac{k\ell_2}{\ell_o}$

(C) $\frac{k\ell_0}{\ell_2}$ and $\frac{k\ell_0}{\ell_1}$ (D) $\frac{k\ell_2}{\ell_0}$ and $\frac{k\ell_1}{\ell_0}$

Q21. Reading of spring balance is (g = 10 m/s^2):-



(A) 1 kg wt (B) 2 kg wt

(C) 3 kg wt (D) 4 kg wt

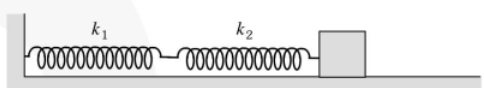
Q22. A block of mass 4kg is suspended through two light spring balances A and B. Then A and B will read respectively :-



(A) 4 kg and zero kg (B) Zero kg and 4 kg

(C) 4 kg and 4 kg (D) 2 kg and 2 kg

Q23. The mass in the figure can slide on a frictionless surface. When the mass is pulled out, spring 1 is stretched a distance x_1 and spring 2 is stretched a distance x_2. The spring constants are k_1 and k_2 respectively. Magnitude of spring force pulling back on the mass is -



(A) k_1x_1

(B) $(k_1x_1 - k_2x_2)]$

(C) $(k_1x_1 + k_2x_2)$

(D) $0.5\ (k_1 + k_2)\ (x_1 + x_2)$

Pseudo Force

Q24. How much Pseudo force will act on a mass of 3 kg as observed from a reference frame of person of 5kg moving with acceleration of 2 m/s^2 in right direction?

(A) 10 N towards left

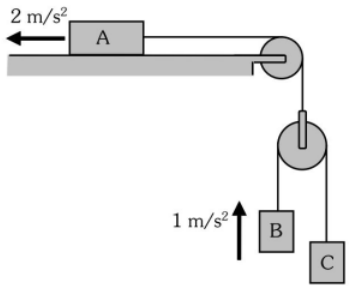
(B) 10 N towards right

(C) 6N towards right

(D) 6N towards left

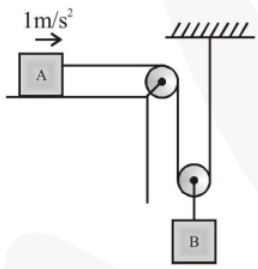
Constraint Motion

Q25. In the setup shown, find acceleration of the block C.



(A) 3 m/s² ↑ (B) 3 m/s² ↓
(C) 5 m/s² ↑ (D) 5 m/s² ↓

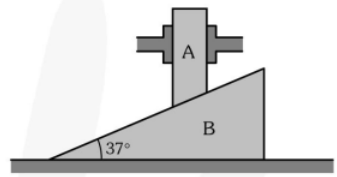
Q26. Find $\vec{a}_B$.



(A) 2 m/s² ↓ (B) $\frac{1}{2}$ m/s² ↑
(C) $\frac{1}{2}$ m/s² ↓ (D) 1 m/s² ↓

Wedge Constraint

Q27. Rod A can slide in vertical direction pushing the triangular wedge B towards left. The wedge is moving toward right with uniform acceleration 4 m/s². Acceleration of the rod A is:



(A) 8 m/s² (B) 4 m/s²
(C) 3 m/s² (D) 2 m/s²

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A particle stays at rest as seen from a frame. Which of the following statements is/are correct:-

(A) The frame is inertial

(B) The frame may be inertial but the resultant force on the particle is zero

(C) The frame may be non-inertial but there is a non-zero resultant force on the partick

(D) The frame is non-inertial

Q2. Consider a block suspended from a light string as shown in the figure. Which of the following pairs of forces constitute Newton's third law pair?



(A) Force with which string pulls on the ceiling and the force with which string pulls on block

(B) Force with which string pulls on the block and weight of the block

(C) Force acting on block due to the earth and force the block exerts on the earth

(D) Force with which block pulls on string and force with which the string pulls on the block

Q3. If a horizontal support exerts an upward force of 10 N on a block of weight 9.8 N placed on it, which of the following statements is/are correct. Assume acceleration due to gravity to be 9.8 m/s^2.

(A) The block exerts a force of 10 N on the support.

(B) The block exerts a force of 9.8 N on the support.

(C) The block has an upward acceleration.

(D) The block has a downward acceleration.

Q4. A man of mass m is on a ladder of mass M hanging from fixed support vertically. Choose the correct statement/statements about the situation.

(A) If the man climbs up the ladder with decreasing speed, force applied by ladder on the support is less than (m+M)g.

(B) If the man climbs up the ladder with uniform speed, force applied by ladder on the support is equal to (M+m)g.

(C) If the man climbs down the ladder with increasing speed, force applied by ladder on the support is more than (M+m)g.

(D) If the man climbs down the ladder with decreasing speed, force applied by ladder on the support is less than (M+m)g.

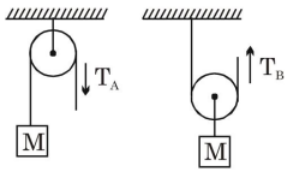
SUBJECTIVE TYPE

Q5. A force F applied to an object of mass m_1 produces an acceleration of 3.00 m/s^2. The same force applied to a second object of mass m_2 produces an acceleration of 1.00 m/s^2.

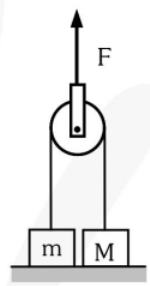
(i) What is the value of the ratio m_1 / m_2 ?

(ii) If m_1 and m_2 are combined, find their acceleration under the action of the force F.

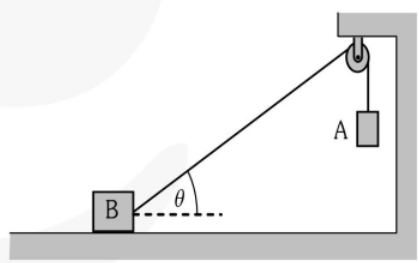
Q6. Consider the two configurations shown in equilibrium. Find ratio of T_A/T_B. (Ignore the mass of the rope and the pulley)



Q7. In the system shown, pulley and strings are ideal. The vertically upward pull F is being increased gradually, find magnitude of F and acceleration of the 5 kg block at the moment the 10 kg block leaves the floor. (m = 5 kg, M = 10 kg)



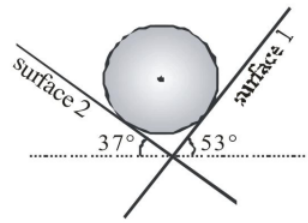
Q8. The block A is moving downward with constant velocity v_0. Find the velocity of the block B, when the string makes an angle θ with the horizontal.



MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. A sphere of mass 10 kg is placed in equilibrium in a V shaped groove plane made of two smooth surfaces 1 and 2 as shown in figure. ($g = 10\ ms^{-2}$)

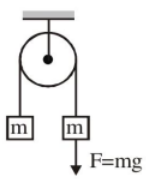
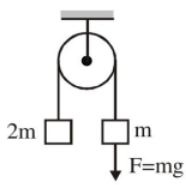
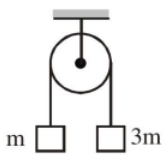


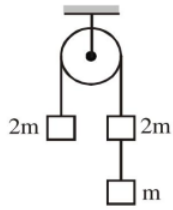
Column I	Column II
(A) Normal reaction by Surface 1	(P) Zero
(B) Normal reaction by surface 2	(Q) 60 N
(C) Force on sphere by Earth	(R) 80 N
(D) Net force on sphere	(S) 100 N
	(T) 120 N

Q10. Velocity of three particles A, B and C varies with time t as, $\vec{v}_A = (2t\hat{i} + 6\hat{j})m/s$; $\vec{v}_B = (3\hat{i} + 4\hat{j})m/s$ and $\vec{v}_C = (6\hat{i} - 4t\hat{j})$ m/s. Regarding the pseudo force match the following table :

Column I	Column II
(A) On A as observed by B	(P) Along positive x–direction
(B) On B as observed by C	(Q) Along negative x–direction
(C) On A as observed by C	(R) Along positive y–direction
(D) On C as observed by A	(S) Along negative y–direction
	(T) Zero

Q11. Match the situations in column I to the accelerations of blocks in the column II (acceleration due to gravity is g and F is an additional force applied to one of the blocks ?

Column I	Column II
(A) 	(P) $\frac{g}{5}$
(B) 	(Q) $\frac{g}{3}$
(C) 	(R) $\frac{g}{2}$

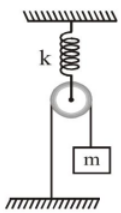
(D) 

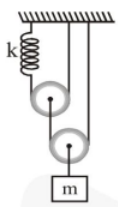
(S) $\frac{2g}{3}$

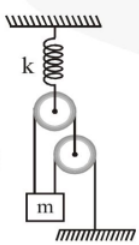
(T) zero

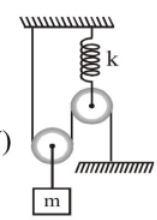
Q12. Find the extension in equilibrium in column-I and match them with column-II and choose the correct option from codes given belowl :

Column-I

(I) 

(II) 

(III) 

(IV) 

Column-II

(P) $\frac{mg}{k}$

(Q) $\frac{4mg}{3k}$

(R) $\frac{2mg}{k}$

(S) $\frac{mg}{4k}$

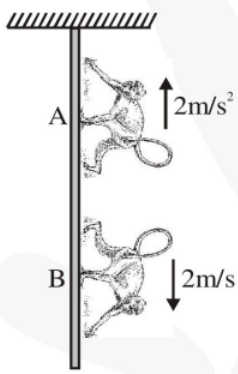
Codes :

	I	II	III	IV
(A)	R	Q	S	P
(B)	R	S	Q	P
(C)	P	R	S	Q
(D)	Q	P	R	S

SINGLE CORRECT TYPE

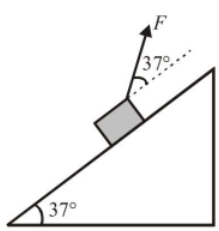
Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Two monkeys of masses 10 kg and 8 kg are moving along a vertical light rope, the former climbing up with an acceleration of 2 m/s^2, while the latter coming down with a uniform velocity of 2 m/s. Find tension in the rope at the fixed support.



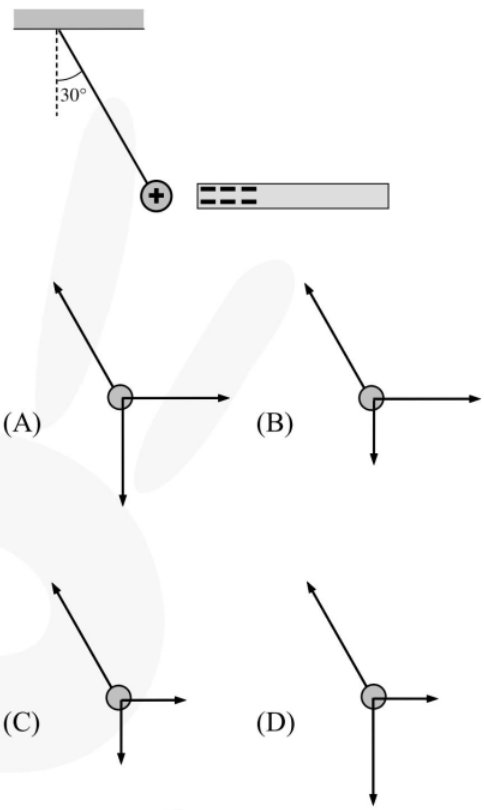
(A) 180 N (B) 200 N
(C) 80 N (D) 216 N

Q2. A block resting on a smooth inclined plane is acted upon by a force F as shown. If mass of block is 2 kg and F = 20 N and sin 37°= 3/5, the acceleration of block is -



(A) 2 m/s^2 (B) 6 m/s^2
(C) 8 m/s^2 (D) zero

Q3. A small electrically charged sphere is suspended vertically from a thread. An oppositely charged rod is brought close to the sphere such that the sphere is in equilibrium displaced from the vertical by an angle of 30°. Which one of the following best represents the free body diagram for the sphere?

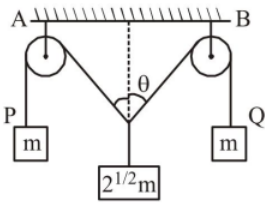


Q4. A flexible chain of weight W hangs between two fixed points A & B which are at the same horizontal level. The inclination of the chain with the horizontal at both the points of support is θ. What is the tension of the chain at the mid point?

(A) $\frac{W}{2}$. *cosec* θ (B) $\frac{W}{2}$. *tan* θ

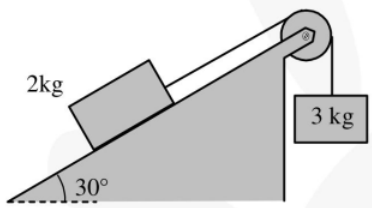
(C) $\frac{W}{2}$ *cot* θ (D) none

Q5. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be :



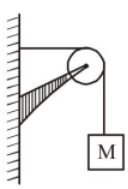
(A) 0° (B) 30°
(C) 45° (D) 60°

Q6. In the arrangement shown, the 2 kg block is held to keep the system at rest. The string and pulley are ideal. When the 2 kg block is set free, by what amount the tension in the string changes? [g = 10 m/s²]



(A) Increase of 12 N (B) Decrease of 12 N
(C) Increase of 18 N (D) Decrease of 18 N

Q7. A string of negligible mass going over a clamped pulley of mass *m* supports a block of mass *M* as shown in the figure. The force on the pulley by the clamp is given :



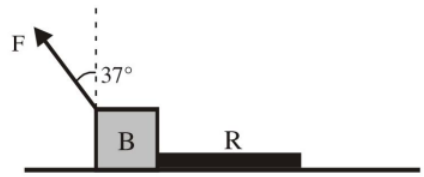
(A) $\sqrt{2}\,Mg$

(B) $\sqrt{2}\,mg$

(C) $\sqrt{(M+m)^2+m^2}\ g$

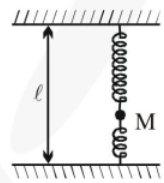
(D) $\sqrt{(M+m)^2+M^2}\ g$

Q8. A block B is tied to one end of a uniform rope R as shown. The mass of block is 2 kg and that of rope is 1 kg. A force F = 15 N is applied at angle 37° with vertical. The tension at the mid-point of rope is -



(A) 1.5 N (B) 2 N
(C) 3N (D) 4.5 N

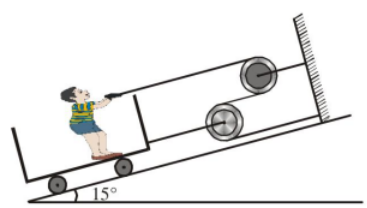
Q9. A small ball of mass M is held in equilibrium with two identical springs as shown in the figure. Force constant of each spring is k and relaxed length of each spring is $\ell/2$. What is distance between the ball and roof ?



(A) $\frac{\ell}{2}+\frac{Mg}{k}$ (B) $\frac{\ell}{2}-\frac{Mg}{k}$

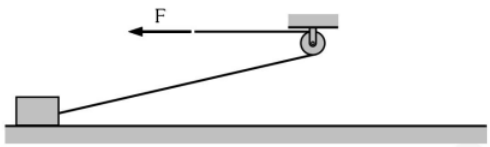
(C) $\frac{\ell}{2}+\frac{Mg}{2k}$ (D) $\frac{\ell}{2}-\frac{Mg}{2k}$

Q10. A trolley is being pulled up an incline plane by a man sitting on it (as shown in figure). He applies a force of 250 N. If the combined mass of the man and trolley is 100 kg, the acceleration of the trolley will be :
[sin15° = 0.26]



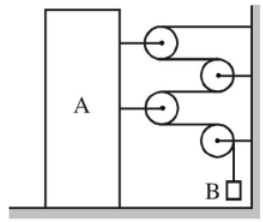
(A) 2.4 m/s² (B) 9.4 m/s²
(C) 6.9 m/s² (D) 4.9 m/s²

Q11. A heavy cart is pulled by a constant force F along a smooth horizontal track with the help of a rope that passes over a fixed pulley, as shown in the figure. Assume the tension in the rope remain constant and consider motion of the cart until it reaches vertically below the pulley. As the cart moves to the right, its acceleration



(A) decreases.
(B) increases.
(C) remains constant.
(D) is zero

Q12. Block A is moving away from the wall at a speed v and acceleration a.



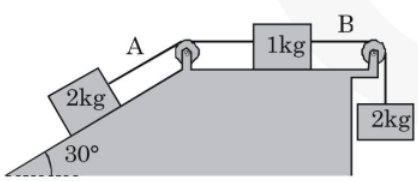
(A) Velocity of B is v with respect to A.
(B) Acceleration of B is a with respect to A.
(C) Acceleration of B is 4a with respect to A.
(D) Acceleration of B is $\sqrt{17}$a with respect to A.

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. System shown in the figure is released from rest. [g = 10 m/s^2]
Which of the following statements is (are) true?



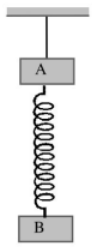
(A) Magnitude of acceleration of each block is 2 m/s^2.
(B) Tension in the cord A is more than that in cord B.
(C) Tension in the cord A is less than that in cord B.
(D) Magnitude of difference in tensions in both the cords is 2 N.

Q2. A carpenter of mass 50 kg is standing on a weighing machine placed in a lift of mass 20 kg. A light string is attached to the lift. The string passes over a smooth pulley and the other end is held by the carpenter as shown. When carpenter keeps the lift moving upward with constant velocity (g = 10 m/s^2)



(A) the reading of weighing machine is 15 kg
(B) the man applies a force of 350 N on the string
(C) net force on the man is 150 N
(D) Net force on the weighing machine is 150 N

Q3. In the given figure both the blocks have equal mass. When the thread is cut, which of the following statements give correct description immediately after the thread is cut?



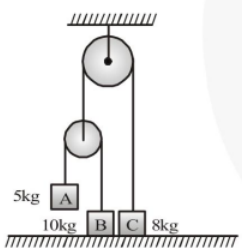
(A) Relative to the block A, acceleration of block B is 2g upwards.

(B) Relative to the block B, acceleration of block A is 2g downwards.

(C) Relative to the ground, accelerations of the blocks A and B are both g downwards.

(D) Relative to the ground, accelerations of the blocks A and B are 2g downwards and zero respectively.

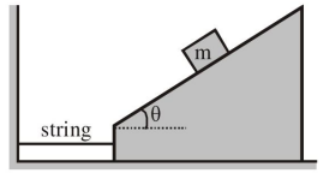
Q4. In the following arrangement the system is initially at rest. The 5 kg block is now released. Assuming the pulleys and string to be massless and smooth, the acceleration of blocks is [g = 10 m/s^2]



(A) $a_A = \frac{g}{7}$ (B) $a_B = 0$ m/s^2

(C) $a_C = \frac{5}{7}$ m/s^2 (D) $2a_C = a_A$

Q5. Refer the system shown in the figure. Block is sliding down the wedge. All surfaces are frictionless. Find correct statement(s) :



(A) Acceleration of block is gsinθ

(B) Acceleration block is gcosθ

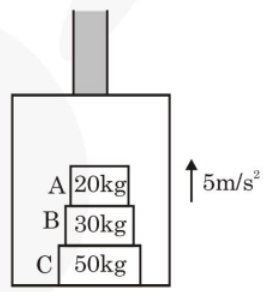
(C) Tension in the string is mgcos^{2}θ

(D) Tension in the string is mgsinθ.cosθ

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II.ONLY ONE of these four options corresponds to a correct matching For each question, choose the option corresponding to the correct matching.*

Q6. In shown situation elevator is moving upward with acceleration of 5 m/s^2.



List-I	List-II
(P) Net force acting on B	(1) 150 N
(Q) Normal reaction between A and B	(2) 300 N
(R) Normal reaction between B and C	(3) 750 N
(S) Normal reaction between C and elevator	(4) 1500 N

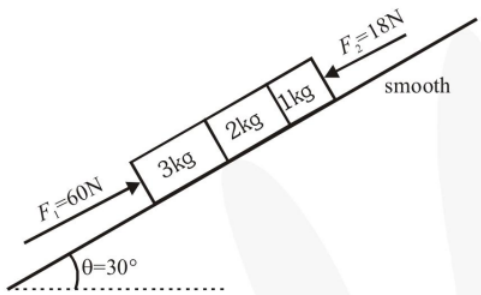
Code :

	P	Q	R	S
(A)	1	2	3	4
(B)	2	4	3	1
(C)	1	2	4	3
(D)	3	2	1	4

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q7. In the diagram shown in figure ($g = 10\ m/s^2$)



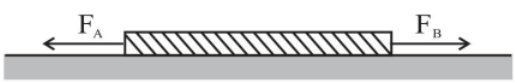
Column I	Column II
(A) Acceleration of 2 kg block in m/s^2	(P) 8
(B) Net force on 3 kg block in newton	(Q) 25
(C) Normal reaction between 2 kg and 1 kg in newton	(R) 2
(D) Normal reaction between 3 kg and 2 kg in newton	(S) 45
	(T) None

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 8 to 11)

A uniform rope of mass (m) and length (L) placed on frictionless horizontal ground is being pulled by two forces F_A and F_B at its ends as shown in the figure. As a result, the rope accelerates toward the right.



Q8. Acceleration (A) of the rope is -

(A) zero

(B) $a = \frac{F_A + F_B}{m}$

(C) $a = \frac{F_A - F_B}{m}$

(D) $a = \frac{F_B - F_A}{m}$

Q9. Tension (T) at the mid point of the rope is -

(A) $T = F_B - F_A$

(B) $T = F_A + F_B$

(C) $T = \frac{1}{2}(F_B - F_A)$

(D) $T = \frac{1}{2}(F_A + F_B)$

Q10. Expression (T_x) of tension at a point at distance x from the end A is -

(A) $T_x = \left(\frac{F_B - F_A}{L}\right)x + F_A$

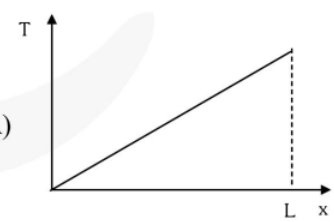
(B) $T_x = \left(\frac{F_B - F_A}{L}\right)x - F_A$

(C) $T_x = \left(\frac{F_B - F_A}{L}\right)x + F_B$

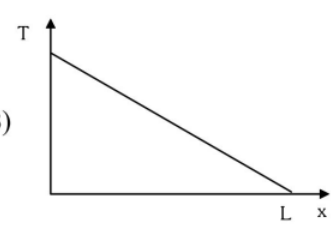
(D) $T_x = \left(\frac{F_B - F_A}{L}\right)x - F_B$

Q11. Which of the following graph best represents variation in tension at a point on the rope with distance x of the point from the end A?

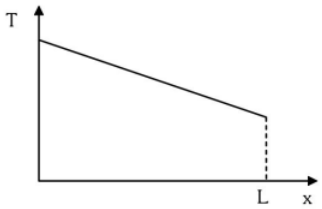
(A)



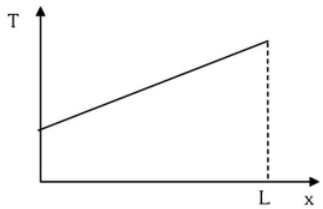
(B)



(B)

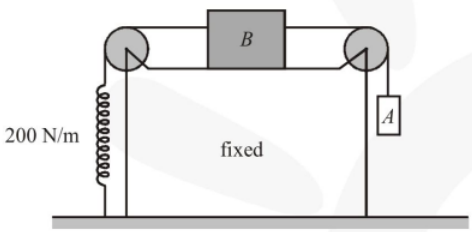


(D)



Paragraph # 2 (Ques. 12 to 14)

The figure shown blocks A and B are of mass 2 kg and 8 kg and they are connected through strings to a spring connected to ground. The blocks are in equilibrium. ($g = 10 m/s^2$)



Q12. The elongation of the spring is -

(A) 1 cm (B) 10 cm

(C) 0.1 cm (D) 1m

Q13. Now the block A is pulled downwards by a force gradually increasing to 20 N. The new elongation of spring is -

(A) 2 cm (B) 4 cm

(C) 20 cm (D) 40 cm

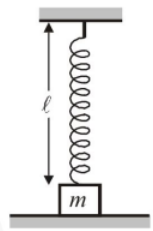
Q14. Now the force on A is suddenly removed. The acceleration of block B becomes -

(A) 1.0 m/s (B) 2.0 m/s^2

(C) 3.0 m/s^2 (D) 4.0 m/s^2

Paragraph # 3 (Ques. 15 to 17)

A block of mass m is placed on a smooth horizontal floor is attached to one end of spring. The other end of the spring is attached to fixed support. When spring is vertical it is relaxed. Now the block is pulled towards right by a force F, which is being increased gradually. When the spring makes angle 53° with the vertical, block leaves the floor.



Q15. When blocks leaves the table, the normal force on it from table is -

(A) mg (B) zero

(C) $\frac{4mg}{3}$ (D) $\frac{3mg}{4}$

Q16. Force constant of the spring is -

(A) $\frac{5mg}{2\ell}$ (B) $\frac{15mg}{8\ell}$

(C) $\frac{5mg}{3\ell}$ (D) $\frac{5mg}{4\ell}$

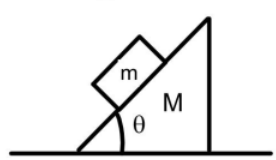
Q17. When the block leaves the table, the force F is -

(A) $\frac{3mg}{4}$ (B) $\frac{4mg}{3}$

(C) $\frac{3mg}{5}$ (D) $\frac{4mg}{5}$

SUBJECTIVE TYPE

Q18. A block of mass m lies on wedge of mass M as shown in figure.



With what minimum acceleration must the wedge be moved towards right horizontally so that block m falls freely.

Answer Key

Ex-1											
1	2	3	4	5	6	7	8	9	10	11	12
B	A	C	B	C	D	B	B	D	D	C	C
13	14	15	16	17	18	19	20	21	22	23	24
B	A	B	B	C	C	B	A	D	C	A	D
25	26	27									
A	C	C									

Ex- 1A

1. B, C **2.** C, D **3.** A, C **4.** A, B

5. (i) $\frac{M_1}{M_2} = \frac{1}{3}$ (ii) a=3/4 m/s^2 **6.** 2 **7.** 200N, 10m/s^2

8. $v_0 / \cos\theta$ **9.** (A)-Q, (B)-R, (C)-S, (D)-P **10.** (A)-T, (B)-R, (C)-R, (D)-Q

11. (A)-R, (B)-T, (C)-R, (D)-P **12.** B

Ex-2											
1	2	3	4	5	6	7	8	9	10	11	12
B	A	D	C	C	B	D	A	C	D	A	D

Ex- 2A

1. A,C,D **2.** A,B **3.** A,B,D **4.** A,B,C,D **5.** A,D

6. A **7.** (A)-R, (B)-T, (C)-Q, (D)-T **8.** D **9.** D

10. A **11.** D **12.** B **13.** C **14.** B

15. B **16.** A **17.** B **18.** a = g cotθ

-: Important Notes & Formulas :-